
Dynamic Dragon: Integrating Regime Detection into Strategic Asset Allocation

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Abstract

Conventional allocations—such as 60/40 and risk parity—have excelled in recent decades, but can suffer painful drawdowns during high-inflation episodes or periods of sustained positive stock–bond correlation. The *Dragon Portfolio*, developed by Artemis Capital Management, seeks robustness by incorporating defensive assets—gold, commodity trend-following, and active long-volatility strategies—into the core allocation. I extend this framework by replacing fixed allocations with a price-implied, regime-adaptive overlay. Using price-based features across major asset classes, I identify five market regimes with a Gaussian mixture model (GMM) and assign each regime a predefined portfolio tilt that has historically been most favorable in that environment. In-sample (1927–2019), the overlay improves long-horizon risk-adjusted performance while maintaining similar volatility and drawdowns; out-of-sample (2020–2024) results are comparable. Overall, this regime-adaptive approach produces transparent, implementable rules that require no macroeconomic forecasts, and offer institutional allocators a robust, data-driven framework for strategic asset allocation.

“Every investment has an ideal environment in which it flourishes. I know that there are good and bad environments for all asset classes. And I know that in one’s lifetime, there will be a ruinous environment for one of those asset classes.”

– Ray Dalio, Founder of Bridgewater Associates

1 Introduction

In their whitepaper *The Allegory of the Hawk and the Serpent* [1], Artemis Capital Management highlighted the limitations of conventional asset allocation strategies, such as the classic 60/40 equity–bond and risk parity portfolio. By analyzing nearly a century of empirical data dating back to the 1920s, their study evaluated the long-term performance of various portfolio allocation strategies across a range of macroeconomic regimes. While traditional portfolios have delivered strong returns during periods of secular growth, particularly in recent decades, they have proven vulnerable during episodes of high inflation or market stress. This weakness was evident in 2022, when fixed income assets not only failed to provide adequate protection against equity drawdowns, but more concerningly, exhibited rising positive correlation with risk assets amid a low-interest-rate and rising-inflation environment. To address the challenge of growing and preserving capital through both secular booms and busts, Artemis introduced the *Dragon Portfolio*, an aggressively diversified allocation consisting of five asset classes: equities, fixed income, gold, commodity trend-following, and active long-volatility strategies. This portfolio not only outperformed conventional portfolios over the long term but also demonstrated greater resilience during market shocks and prolonged bear markets.

While the Dragon Portfolio offers a compelling framework for robust long-term strategic asset allocation, its intentionally static design does not reflect the dynamic and adaptive approach of most institutional asset allocators. In practice, institutions reassess risk premia, correlations, and volatility on an ongoing basis and rebalance exposures as market conditions, policy regimes, and business cycles evolve.

This paper develops a dynamic extension of the Dragon Portfolio. Inspired by Two Sigma’s unsupervised regime modeling study [2], I construct a Gaussian mixture model using price-based features across major asset classes to identify and characterize distinct market regimes over the past century. For each regime, I evaluate the optimal asset allocation and prescribe a portfolio tilt that replaces the original static weights. The objective is to preserve the Dragon Portfolio’s structural diversification while enhancing its performance through improved adaptability to evolving market environment.

2 Literature Review

The detection and modeling of market regimes has long been a subject of interest across both academic research and the investment industry. The core objective is to identify latent structural breaks and patterns in asset returns, volatility and macroeconomic indicators, and to integrate these insights into more adaptive portfolio strategies.

Early contributions came primarily from economists using econometric methods. A foundational approach was the Markov Switching Model [4], introduced by Hamilton in 1989, which models time series as mixtures of unobserved states governed by a Markov process. This framework enabled dynamic modeling of macroeconomic time series, particularly GDP growth, under the assumption that economic states alternate between discrete regimes with distinct statistical properties. Hamilton’s work laid the groundwork for incorporating regime-switching behavior into financial return models and asset allocation strategies.

Building on this foundation, Ang and Bekaert extended regime-switching frameworks by examining how expected returns, volatilities, and cross-asset correlations evolve across different market regimes [5]. They demonstrated that bear markets are not only characterized by lower expected returns but also by heightened volatility and increased co-movement among global assets—undermining the traditional benefits of diversification. Their regime-aware global CAPM showed that conditioning portfolio decisions on regime probabilities leads to superior performance relative to static allocation strategies. Jun Tu further supported these findings, showing that incorporating regime-switching into equity portfolios could enhance annualized returns by approximately 2%, even after accounting for transaction costs and estimation uncertainty [6].

In recent years, the field has increasingly embraced machine learning techniques to improve the responsiveness, scalability, and predictive power of regime detection models. Firms such as AQR, Two Sigma, and Bridgewater have incorporated tools such as clustering algorithms, feature engineering, and deep learning into their regime frameworks. A notable example is Two Sigma’s use of a Gaussian mixture model to identify distinct market environments. This unsupervised, data-driven approach enables the discovery of previously unrecognized or unintuitive patterns in financial behavior. The result is a more flexible and empirically grounded framework capable of capturing complex regime dynamics in real time.

Empirical evidence consistently shows that regime-aware strategies improve risk-adjusted returns and mitigate drawdowns by aligning portfolio exposure with prevailing economic conditions. Evolving body of research and practitioner experimentation continues to push the field toward more robust, interpretable, and forward-looking regime detection tools, which serve as an important component in modern strategic asset allocation.

3 Hypotheses

Hypothesis #1: Without prior knowledge of macroeconomic conditions, clustering models reveal latent market regimes that intuitively align with prevailing narratives about the market environment at the time.

I test whether clustering models can uncover meaningful and interpretable market regimes that correspond to widely accepted macroeconomic and financial market conditions over the past century. If the identified clusters align with historically recognized episodes, such as the stagflation of the 1970s, the liquidity-driven

expansion of the 2000s, or the COVID-era crisis, it would support the premise that financial markets behave distinctly under different regimes, and these regimes can be inferred directly from asset return patterns.

Hypothesis #2: Market regimes persist for extended periods, often spanning years or even decades.

Market regimes should not be random or short-lived, but instead reflect slow-moving economic and policy dynamics, such as demographic trends, monetary policy stances, and credit cycles. Demonstrating regime persistence would support the viability of integrating regime signals into dynamic asset allocation frameworks, and enable meaningful portfolio adjustments without excessive turnover.

Hypothesis #3: A regime-aware Dragon Portfolio improves long-horizon risk-adjusted performance relative to the static Dragon benchmark.

By shifting weights toward assets that historically outperform in a given regime—and away from those that lag—the regime-adaptive overlay should raise risk-adjusted returns over extended horizons. I evaluate this using annualized returns, Sharpe ratio, and related performance metrics.

Hypothesis #4: A regime-aware Dragon Portfolio reduces drawdowns and improves resilience relative to the static Dragon benchmark.

Because static weights can be ill-suited to prolonged stress or structural change, a regime-adaptive overlay should mitigate peak-to-trough losses and hasten recoveries. I assess this using maximum drawdown, average drawdown, and time to recovery.

4 Objectives, Constraints, and Benchmarks

4.1 Objectives

The primary objective of this strategy is to develop a dynamic, regime-aware extension of the Dragon Portfolio that enhances long-term performance while preserving its core principle of structural diversification. The strategy aims to first identify the prevailing market regime using a clustering-based model and adjusting asset allocations accordingly. By adapting to changing market environments, the strategy aims to improve risk-adjusted returns and reduce drawdowns relative to the original static Dragon Portfolio.

4.2 Constraints

To ensure the strategy remains realistic and aligned with institutional asset management practices, several practical constraints are applied:

- **Asset Class Inclusion:** The portfolio is restricted to the five asset categories defined in the original Dragon Portfolio: equities, fixed income, gold, commodity trend-following, and active long volatility strategies. This constraint allows for a meaningful comparison with its static counterpart.
- **Rebalancing Frequency:** The portfolio may be rebalanced no more than once per year. This reflects the frequency of strategic asset allocation reviews typically conducted by institutional investors. Tactical shifts may occur more frequently in practice, but this strategy is intended to remain low-turnover and grounded in long-term regime dynamics.
- **Allocation Adjustment Range:** To limit excessive turnover and maintain portfolio stability, each asset class is allowed to adjust its weight by a maximum of 5 percentage points up or down relative to its baseline allocation. This constraint ensures that regime-driven changes remain incremental and controlled.

4.3 Benchmarks

The static Dragon Portfolio serves as the benchmark for evaluating the performance of the dynamic regime-aware strategy. Comparative analysis will be conducted across both the training period (1927–2019) and the out-of-sample test period (2020–2024). Evaluation metrics include annualized return, volatility, Sharpe ratio, and maximum drawdown.

5 Market Regime Clustering Analysis

Assessing portfolio behavior across environments first requires defining those market regimes. Regimes represent periods during which asset returns, volatilities, and cross-asset correlations exhibit persistent and distinct patterns. Clear regime definitions are foundational in building adaptive asset allocation strategies.

Traditionally, regimes have been labeled by experts using macroeconomic indicators, policy shifts, or historical narratives, such as labeling periods as “booms and busts” or “risk-on/risk-off”. Such approaches are inherently subjective, susceptible to hindsight bias, and can miss latent structural changes that are not captured by conventional narratives.

I instead adopt a data-driven approach that infers regimes directly from market behavior. The premise is that co-movements in asset prices contain information about the underlying state of the world, even when that state is not directly observed. Concretely, I estimate a Gaussian mixture model on price-based features (returns and pairwise correlations) to identify a finite set of latent regimes. These statistically grounded regime estimates form the basis for the regime-aware portfolio construction that follows.

5.1 Data Descriptive Analysis and Feature Construction

I assemble a cross-asset panel spanning equities, bonds, commodities, and derived factors. These asset classes can serve as proxies for specific macroeconomic conditions: equities reflect expectations about economic growth and corporate profitability; Treasury bonds capture monetary policies and inflation expectations; gold and crude oil respond to inflation pressures; and emerging market equities reflect global risk appetite and capital flows.

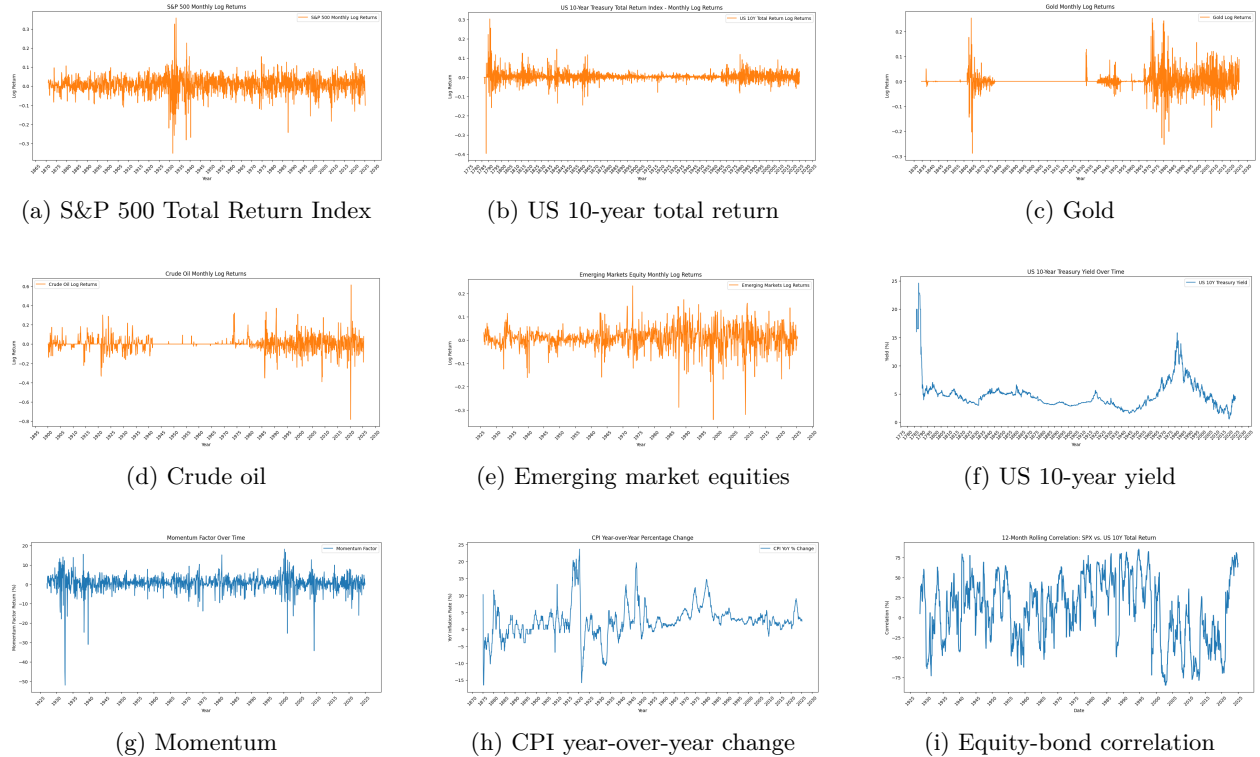


Figure 1: Descriptive analysis of the features

Most features are monthly log returns (shown as the orange series in the figures). I also include level variables—such as the 10-year U.S. Treasury yield and the momentum factor from the Ken French Data Library

(displayed in blue). These level indicators are intended to capture slower-moving structural conditions, particularly those associated with policy regimes and trend persistence.

To ensure that the feature set captures complementary information rather than redundant signals, I analyzed the pairwise correlations between all input features. This step helps verify that the selected assets and factors have little overlapping exposures. As shown in the correlation heatmap below, most features exhibit low to moderate correlations. This supports their inclusion as independent inputs into the clustering model.

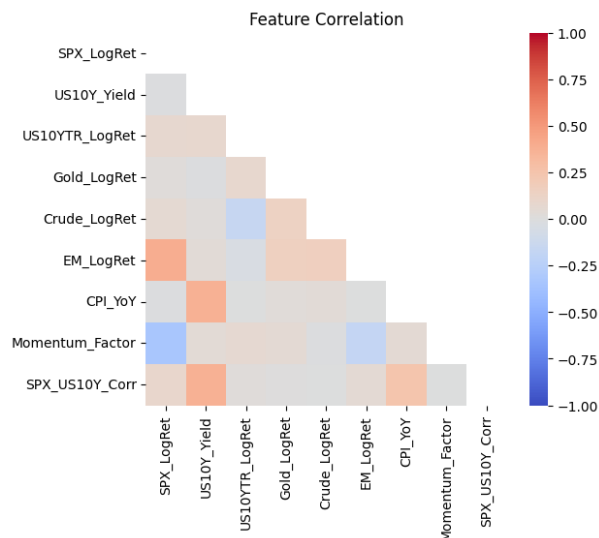


Figure 2: Correlation of selected features

The final feature set includes:

- **Equities:** S&P 500 Total Return Index, MSCI Emerging Markets Equity Index
- **Fixed income:** U.S. 10-Year Treasury Bonds
- **Real assets:** Gold and Crude Oil
- **Derived market dynamics factors:** Momentum factor (from Ken French's data library) and the rolling correlation between equities and bonds

5.2 Model Selection and Implementation

Following the methodology presented in the Two Sigma whitepaper, I use a Gaussian mixture model (GMM) to group time periods based on similarities in asset return characteristics. GMM offers greater flexibility than commonly used clustering algorithms like K-Means, which assign hard cluster memberships based solely on distance metrics.

GMM is particularly well-suited for financial market analysis, as asset returns often exhibit skewness and complex interdependencies. By modeling the full covariance structure within each cluster, GMM can capture the joint behavior of asset returns more accurately, resulting in more realistic and nuanced regime boundaries. Moreover, its probabilistic framework allows for soft classification and quantifies the uncertainty associated with each regime assignment. This is an important consideration for real-world investment decision-making.

After experimenting with various values for the number of clusters, I selected five as the optimal choice based on a combination of statistical model selection criteria and economic interpretability. A detailed discussion of how this decision was made is provided in the following subsection. This choice finds the balance between capturing meaningful variation in market behavior and avoiding overfitting. The resulting clusters distinguish clearly between periods in financial history, each exhibiting distinct return characteristics across asset classes. The time series plot below visualizes the historical regime assignments generated by the model.

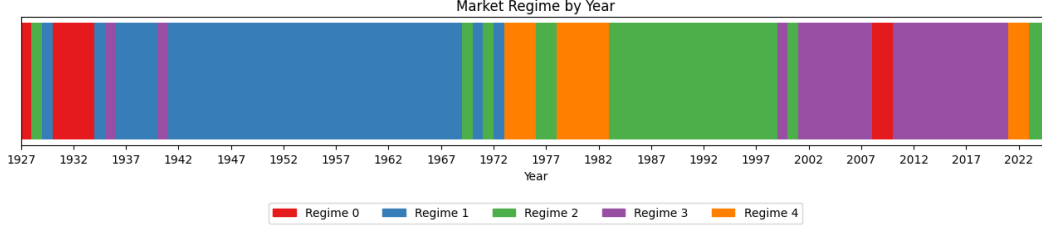


Figure 3: Market regimes

At first glance, the regimes identified by the model resembles those defined in the original Artemis paper, particularly in the timing of major crises, inflationary episodes, and periods of secular growth. However, there are also notable differences. In the next subsection, I examine the characteristics of each identified regime and assess how well they align with widely recognized macroeconomic environments.

5.3 Regime Characteristics and Interpretation

To characterize each regime, I compute the average values of all input features, along with several additional asset classes not included in the clustering model. The results, visualized in the plot below, provide insight into how different assets tend to behave across distinct market environments.

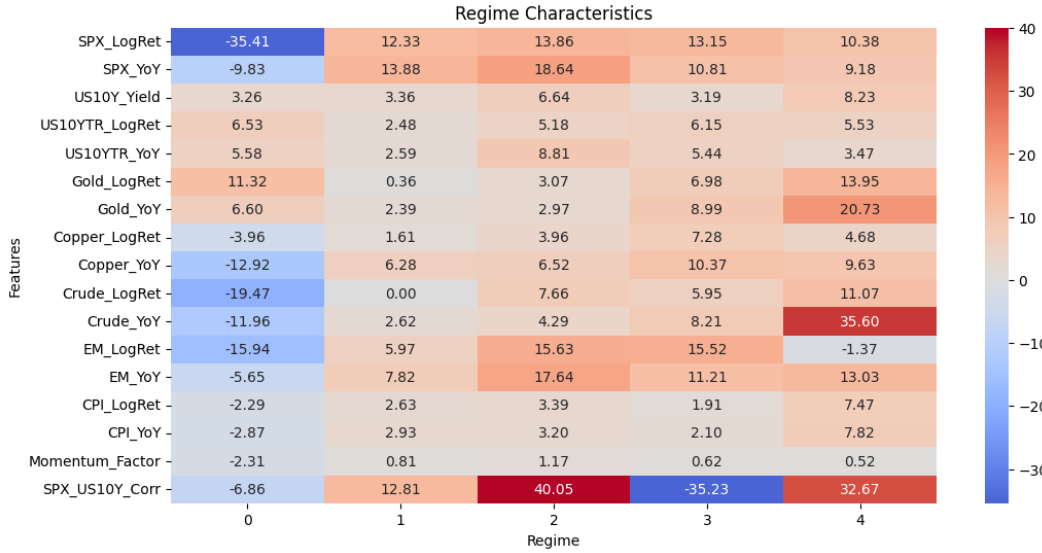


Figure 4: Average values of the features

The five regimes exhibit the following characteristics:

- **Regime 0 (Crisis & Deflation):** Marked by large negative returns in equities and commodities, alongside strong performance in Treasury bonds. This regime aligns with classic crisis or deflationary periods, such as the Great Depression and the 2008 global financial crisis.
- **Regime 1 (Post-War Expansion):** Characterized by broad-based strength across equities, bonds, and commodities. This regime reflects the industrial boom of the 1950s and 1960s, supported by robust productivity growth and stable inflation.
- **Regime 2 (Reaganomics & Tech Boom):** Dominated by strong both equity and bond performance, and positive correlation between equities and bond. This regime captures periods of strong fiscal spending and technological advancement drives market optimism.

- **Regime 3 (Credit expansion & QE):** Characterized by sustained gains in both U.S. and emerging market equities, along with a negative correlation between equities and bonds. This regime reflects a shift in Fed policy following the Greenspan era with aggressive QE and increased market intervention. It is associated with ultra-low interest rates and strong performance across equities and real assets.
- **Regime 4 (Inflation spike & Stagflation):** Defined by poor performance in nominal bonds and strong gains in commodities and gold. This regime aligns with inflationary shocks and stagflationary periods, such as the 1970s oil crisis and the 2021–2022 post-pandemic inflation surge.

I compare the regimes with prevailing narratives in other literature about historical market environments and structural shifts. The inflationary regime aligns closely with major inflation episodes highlighted in the White House Archives, including the 1970s oil shocks and the post-pandemic surge in 2021.

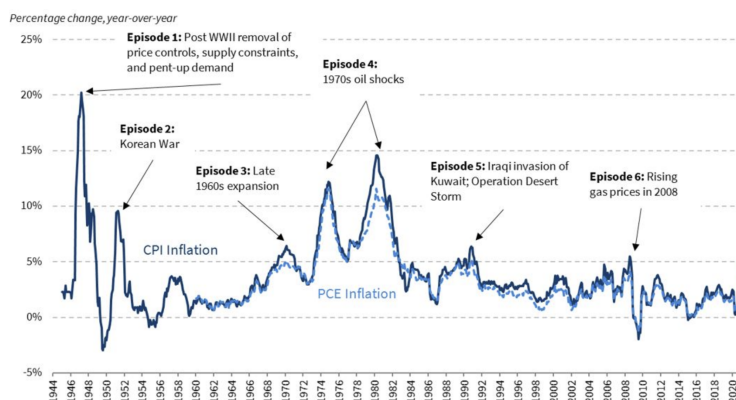


Figure 5: Six episodes of post-WWII inflation (Source: White House Archives)

Additionally, the observed equity-bond correlation dynamics across regimes are consistent with Schroders' analysis, which highlights a structural shift around the year 2000 following the adoption of quantitative easing and the Fed's market interference. These external validations suggest that the model effectively captures key transitions using a purely data-driven approach.

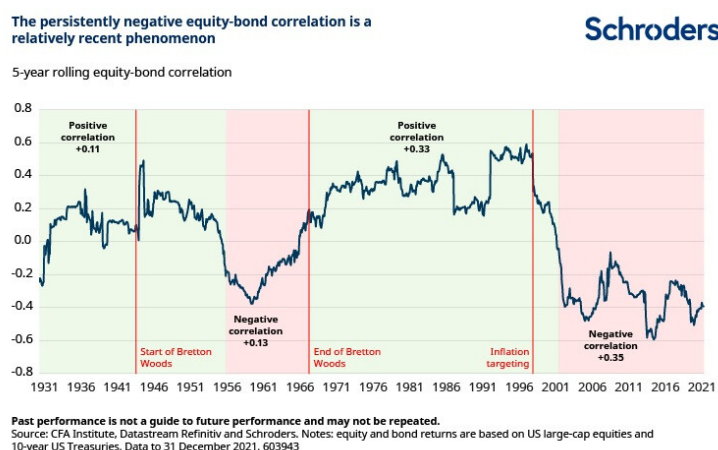


Figure 6: Schroders equity-bond correlation regime analysis

5.4 Evaluating the Optimal Number of Regimes

As previously discussed, the goal of selecting the appropriate number of clusters is to capture meaningful structural differences in market environments without overfitting the data. To determine the optimal number of regimes, I evaluated both quantitative model selection metrics and qualitative economic interpretability.

Statistical Model Selection

I ran the Gaussian mixture models with cluster counts ranging from 1 to 10 and assess model performance using two standard information criteria:

- **Akaike Information Criterion (AIC):** Prioritizes goodness-of-fit and tends to favor more complex models that better explain in-sample variation.
- **Bayesian Information Criterion (BIC):** Penalizes model complexity more heavily and tends to favor simpler models.

Both criteria suggest that model performance improves up to around 4 clusters, after which gains taper off.



Figure 7: Cluster number analysis

Economic Interpretability

In addition to statistical model selection criteria, it is important to qualitatively assess whether the chosen number of clusters adequately captures historically recognized market environments. When comparing the 4 and 5-cluster models, I find that the key breaks are largely consistent. However, the 4-cluster model fails to differentiate between the pre vs post-QE periods, which is an important structural break where the equity-bond correlation shifts from positive to negative. This has significant implications for portfolio construction, since the changing relationship between equities and bonds fundamentally alters diversification dynamics and asset allocation decisions.

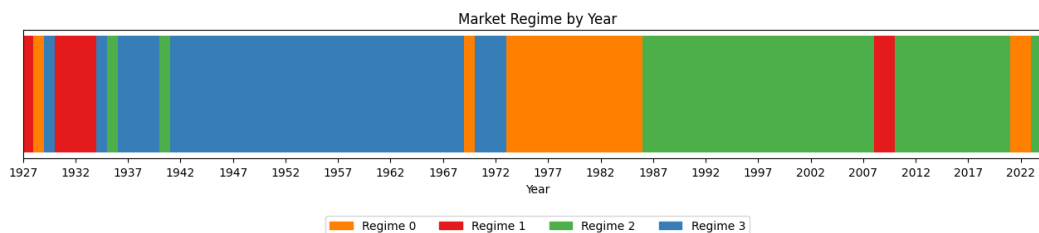


Figure 8: Four-cluster market regimes

6 Integrating Regimes into Asset Allocation Decisions

With the market regimes identified, the next step is to translate that information into actionable investment decisions. This section outlines the framework for incorporating regime classification into portfolio construction. I begin by defining the key components of the strategy: indicators, signals, and rules.

6.1 Indicators

In this context, the indicators are relatively straightforward. They consist of the 9 features selected for the clustering model. These indicators fall into two main categories: return-based and level-based variables.

- **Log returns:** S&P 500 Total Return Index, 10-Year Treasury Total Return Index, gold, crude oil, MSCI Emerging Markets Equity Index
- **Levels:** 10-Year Treasury yield, Fama-French momentum factor, year-over-year CPI change, equity-bond correlation

6.2 Signals

Signals are the regime classification output of the Gaussian mixture model. Each month, the model produces a probability distribution across the latent regimes, representing the likelihood that current market conditions belong to each cluster. For this strategy, I assign the regime with the highest probability as the prevailing market regime for that period.

6.3 Rules

Finally, I establish a set of rules to guide portfolio rebalancing based on the assigned market regime.

The first step involves calculating the average annualized returns of the five Dragon Portfolio asset classes within each regime. As expected, crisis regime tends to be highly unfavorable for equities but favorable for trend-following strategies. The stagflation regime is particularly strong for gold reflecting its role as a hedge against inflation.

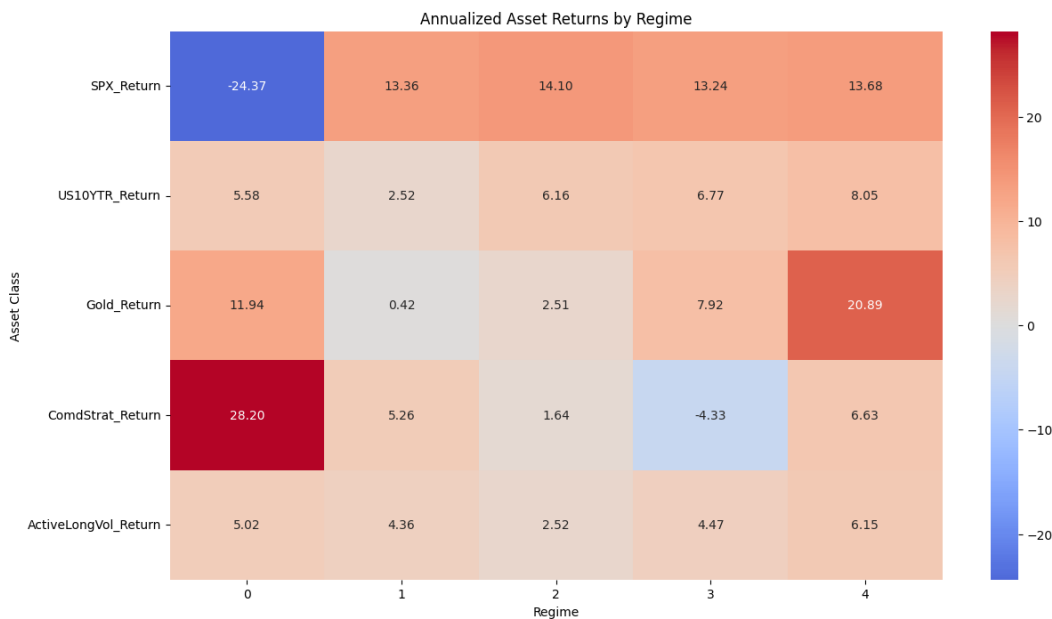


Figure 9: Annualized returns for each asset class and regime

For each asset class, I rank the five regimes based on the asset's average historical performance within each regime, from highest (rank 1) to lowest (rank 5). These rankings provide a regime-specific view of which environments are most favorable or unfavorable for each asset. Using this information, I then adjust the portfolio weights relative to the static Dragon Portfolio, which serves as the baseline allocation. The goal is to overweight assets that tend to outperform and underweight those that historically underperform.

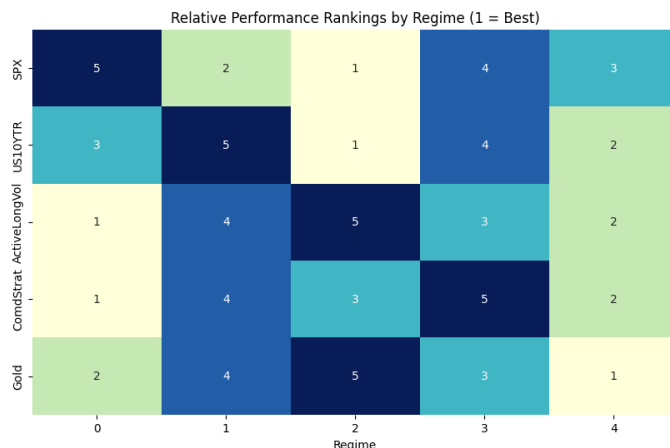


Figure 10: Annualized returns for each asset class and regime

Specifically, if an asset class ranks first in a given regime, meaning it has performed the best in that environment, its portfolio weight is increased by 5% relative to the baseline. If it ranks second, the weight is increased by 2.5%. Conversely, asset classes that rank lowest have their weights reduced by 5%, and those ranking second to last are reduced by 2.5%. After applying these adjustments, all weights are normalized to ensure the portfolio weights summing to 100%. The table below presents the final adjusted weights for each asset class under each identified regime.

Asset class	Regime 0	Regime 1	Regime 2	Regime 3	Regime 4
Equity	17.7	29.5	29.0	23.9	21.3
Bond	16.7	14.4	23.0	17.3	18.2
Gold	20.0	18.3	14.0	21.1	21.3
Commodity trend following	21.4	17.2	18.0	14.4	18.2
Active long volatility	24.2	20.6	16.0	23.3	20.9

Table 1: Assigned Weights by Market Regimes

Additionally, to avoid frequent back-and-forth rebalancing over short time horizons, I made a few manual adjustments to the regime assignments. The plot below displays the final regime assignments used in the analysis.

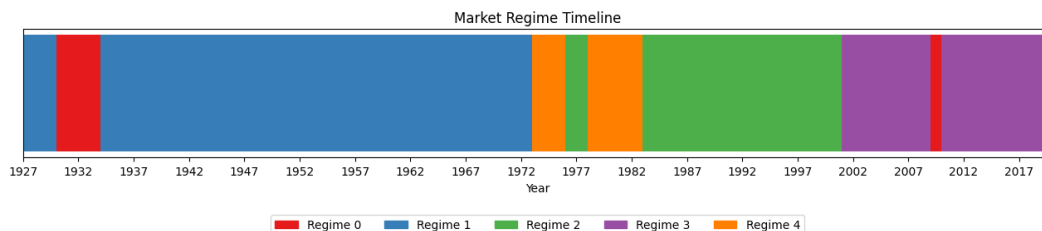


Figure 11: Final market regimes

6.4 Performance Analysis

I can now compare the performance of the static Dragon Portfolio against the regime-adjusted version. The analysis begins with the training dataset, which spans from 1927 to 2019.

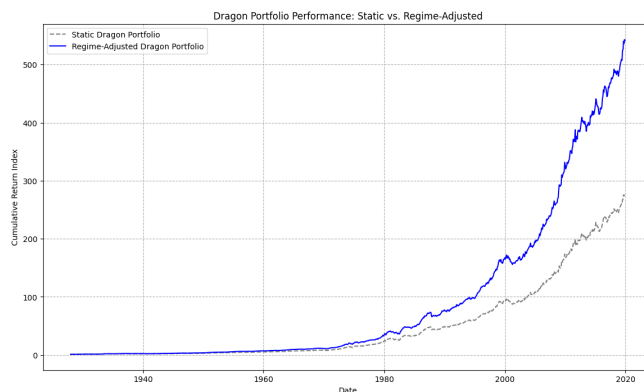


Figure 12: Training period performance

Strategy	Cumulative Return	Annualized Return	Annualized Volatility	Sharpe Ratio	Max Drawdown
Static	27540%	6.35%	6.14%	1.03	-18.01%
Regime-adjusted	54092%	7.14%	6.33%	1.13	-18.34%

Table 2: Performance comparison of regime-based and static Dragon portfolios

The results show that the regime-adjusted Dragon Portfolio delivered nearly twice the cumulative return of its static counterpart over the nearly century-long period. Despite the stronger performance, the portfolio's volatility remained comparable, resulting in a higher Sharpe ratio of 1.13 vs 1.03. However, maximum drawdowns are similar, suggesting regime adjustments do not reduce downside risk. Overall, these suggest that incorporating regime awareness into the Dragon Portfolio enhances long-term performance while maintaining its core principles of structural diversification and resilience.

Now comes the true test: the out-of-sample evaluation using data from 2020 to 2024. This period is interesting because it covers a diverse range of market events, including the Covid shock, a sharp surge in inflation, an aggressive rate-hiking cycle by the Fed, the rare occurrence in 2022 where equities and bonds both suffered significant losses.

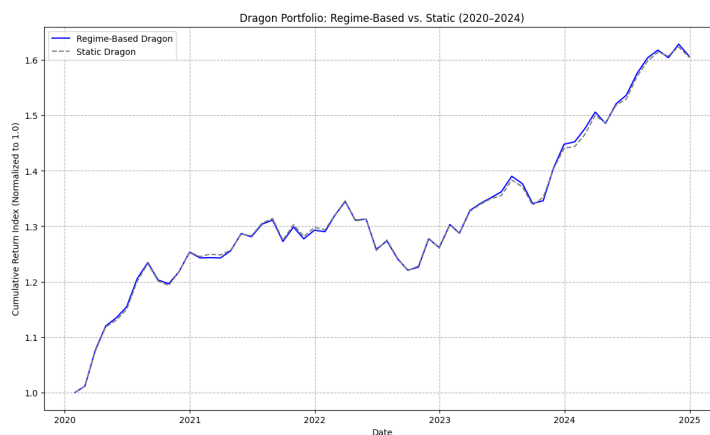


Figure 13: Testing period performance

Strategy	Cumulative Return	Annualized Return	Annualized Volatility	Sharpe Ratio	Max Drawdown
Static	160.49%	9.92%	7.08%	1.40	-9.38%
Regime-adjusted	160.65%	9.94%	7.20%	1.38	-9.21%

Table 3: Performance Comparison of Regime-Based and Static Dragon Portfolios

The out-of-sample results from 2020 to 2024 indicate that the regime-adjusted Dragon Portfolio performed largely in line with its static counterpart, delivering nearly identical cumulative and annualized returns. The dynamic version exhibited slightly higher volatility. This results in a very minor decline in the Sharpe ratio (1.38 vs. 1.40). The regime-adjusted portfolio does show a marginally lower maximum drawdown (-9.21% vs. -9.38%), though the difference is negligible. Overall, these results show limited evidence that the regime-aware strategy meaningfully improved risk-adjusted returns or provided additional downside protection during this period.

6.5 Diagnosing Out-of-Sample Performance

I want to get a deeper understanding of why the regime-aware portfolio’s performance was somewhat underwhelming, and assess whether this was due to potential overfitting in the clustering model or simply the result of an unusually challenging and unpredictable 2020-2024 market environment.

A natural first check is examining the regime classifications assigned to the 2020-2024 period and determine whether they align with the major market events that occurred during those years.

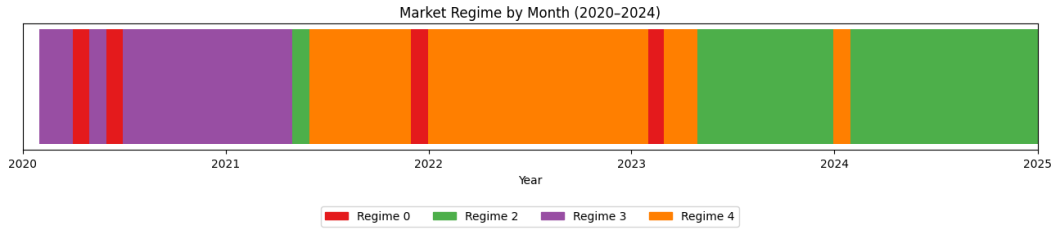


Figure 14: Market regimes from 2020 to 2024

The regime classifications appear to be well-aligned with major market events. The model correctly identified the Covid shock beginning in March 2020 and recognized the rapid recovery that followed. It also captured the inflation surge beginning in mid-2021. Furthermore, the model suggests that we have since entered a regime similar to the post-inflation surge of the 1970s, marked by a return to positive equity-bond correlation. This interpretation is consistent with current market commentary.

Since the regime assignments appear reasonable, I made very minor manual adjustments to avoid excessive rebalancing in consecutive months, helping to reduce unnecessary portfolio turnover.

Date Range	Assigned Regime
January 2020 – April 2021	Regime 3
May 2021 – March 2023	Regime 4
April 2023 – December 2024	Regime 2

Table 4: Manual regime assignments

Next, I evaluated whether the strategy correctly overweighted the asset classes that ultimately performed best during each period. If it failed to do so, this would suggest that the issue lies in the weighting rules rather than the regime classification itself.

Regime	Best Performing Asset	Base (%)	Assigned (%)	Overweighted?
2	Equities	24.0	29.0	Yes
3	Active long vol	21.0	23.3	Yes
4	Commodity Trend following	18.0	18.2	Yes

Table 5: Regime-based asset allocation decisions

The strategy appears to have made the correct asset allocation decisions. To better understand why these decisions did not translate into stronger overall performance, I conduct a detailed performance attribution analysis for each asset class.

Asset Class	Regime 2	Regime 3	Regime 4
Equities	0.0115	-0.0003	-0.0007
Bond	-0.0005	-0.0002	-0.0001
Gold	-0.0086	0.0027	0.0016
Commodity trend following	0.0000	-0.0066	0.0002
Active long vol	-0.0022	0.0065	-0.0000
Total Attribution	0.0002	0.0021	0.0010

Table 6: Performance Attribution by Asset Class and Regime

Overall, total performance attribution from asset allocation decisions is slightly positive across all regimes, suggesting that the strategy generally made sound allocation choices. The primary drag on performance came from underweighting gold in Regime 2 (post-April 2023). Historically, gold underperformed during the post-inflation surge periods of the 1970s and 1980s, which informed the model's expectations. However, in this recent period, gold prices rose significantly (almost 32%).

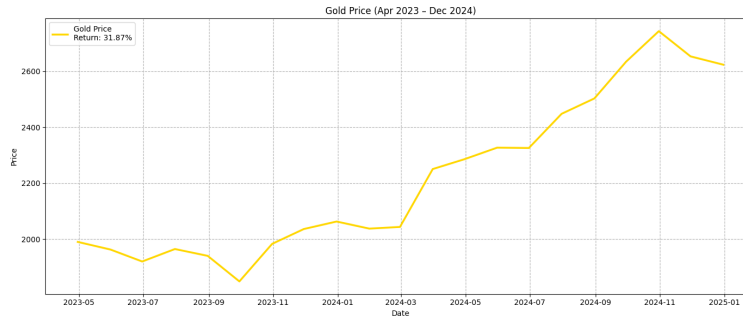


Figure 15: Gold Performance Since April 2023

This is somewhat unusual, as gold typically underperforms in periods of elevated real interest rates. However, the recent environment has been shaped by a unique combination of persistent geopolitical tensions, strong central bank demand (particularly from emerging markets like China) and growing concerns about long-term fiscal sustainability in the US. These factors appear to have supported gold prices despite a rising real rate environment.

This does not prove that the strategy would have clearly outperformed in the absence of this gold anomaly. However, it underscores the importance of flexibility and incorporating expert judgment in strategic asset allocation, particularly for institutional investors navigating environments where historical relationships may temporarily break down.

7 Revisit the hypotheses

Hypothesis #1: Without prior knowledge of macroeconomic conditions, clustering models reveal latent market regimes that intuitively align with prevailing narratives about the market environment at the time.

The results strongly support this hypothesis. Using a Gaussian mixture model trained solely on financial asset performance, the model successfully uncovered regimes that intuitively align with well-known historical market narratives. For instance, the model clearly identified the Great Depression, the stagflation of the 1970s, the QE-driven expansion, and the recent inflation surge in 2021. These clusters were not pre-labeled or guided by economic data, yet the unsupervised learning process captured latent structural shifts in return patterns. This suggests regime detection can be achieved in a purely data-driven fashion.

Hypothesis #2: Market regimes persist for extended periods, often spanning years or even decades.

The results provide strong support for this hypothesis as well. The market regimes identified by the clustering model demonstrate persistence over extended periods, with nearly all lasting for a decade or more. For example, the post-war expansionary regime spans from the 1930s through the 1960s. This suggests that regimes reflect slow-moving structural forces such as demographic shifts, fiscal policy trends, and long-term monetary policy stances. This persistence makes it feasible to incorporate regime signals into strategic asset allocation, as they offer a stable framework over long investment horizons.

Hypothesis #3: A regime-aware Dragon Portfolio improves long-horizon risk-adjusted performance relative to the static Dragon benchmark.

The evidence somewhat supports this hypothesis. Over the training period from 1927 to 2019, the regime-aware strategy outperformed its static counterpart in both absolute and risk-adjusted terms. However, during the out-of-sample test period from 2020 to 2024, performance was largely in line. While not conclusive, these results suggest that incorporating regime awareness into long-term strategic asset allocation has the potential to improve risk-adjusted performance, especially over very long time horizons.

Hypothesis #4: A regime-aware Dragon Portfolio reduces drawdowns and improves resilience relative to the static Dragon benchmark.

These results do not support this hypothesis. In both the training and testing periods, the maximum drawdowns of the regime-aware and static Dragon Portfolios are comparable. While the regime-aware strategy may improve returns over longer horizons, it does not appear to improve downside protection.

8 Discussions and Conclusions

8.1 Rebalancing Rules

I could have explored many alternative rebalancing rules to further optimize the strategy’s performance. However, rebalancing is fundamentally an art in balancing responsiveness and stability. More aggressive rules can boost performance but come at the cost of higher transaction expenses. Smoother rules however may delay adaptation to new market conditions.

More importantly, it is inherently difficult to distinguish between genuine regime shifts and temporary market noise in real time. Sometimes, waiting is the more prudent choice. The purpose of using quantitative techniques like clustering is not to eliminate human judgment, but to inform and enhance it. Ultimately, strategic allocation decisions should reflect the institution’s broader investment mandate, risk tolerance, and governance framework, qualities that no model can fully capture on its own.

8.2 Transaction costs and fees

The performance results presented in this study do not account for transaction costs or performance and management fees associated with hedge fund allocations. However, it is important to recognize that real-world implementation inevitably involves various forms of frictions.

Transaction costs can stem from bid-ask spreads, market impact, and slippage—particularly when reallocating across less liquid asset classes or during periods of elevated market volatility. However, for long-term strategic asset allocations, these costs are typically modest. Empirical studies suggest that transaction costs for highly liquid asset classes generally range from 5 to 15 basis points, with somewhat higher costs associated with alternatives such as trend-following or long volatility strategies. Given the infrequency of rebalancing in this strategy, typically no more than once per decade, the cumulative impact of transaction costs is expected to be minimal over the nearly century-long investment horizon.

8.3 Potential Improvements

Expanding Asset Universe

The current portfolio remains faithful to the original Dragon Portfolio framework, but it is still a simplified representation of the diversified holdings typically found in institutional portfolios. Real-world asset owners often include a broader array of asset classes, such as developed and emerging market equities, inflation-linked bonds, corporate credit, private equity, and real estate. A natural extension of this research would be to evaluate the impact of regime-based adjustments on a representative institutional portfolio and assess whether similar performance enhancements are observed when using a real-world baseline.

Incorporating Additional Features

The field of factor-based investing offers a rich set of variables that could enhance the regime classification process. Beyond the current set of features, additional indicators such as the large and small-cap return spread, the gold-to-copper ratio (often seen as a proxy for economic strength), and valuation metrics like trailing P/E ratios could be incorporated into the clustering model. These features capture different dimensions of market behavior, such as investor sentiment, macroeconomic conditions, and equity market valuations. They may help the model detect more nuanced regime shifts.

Exploring Deep Learning Models

Deep learning might be overhyped now, but it does offer a promising avenue for more advanced regime detection. Techniques such as recurrent neural networks or transformer-based models can capture complex, nonlinear relationships and temporal dependencies in financial data that traditional models may miss. These models can learn latent patterns of market dynamics from much higher dimensional inputs, potentially uncovering more subtle regime structures.

8.4 Conclusions

For as long as I can remember, I’ve been fascinated by asset allocation. The impact of true diversification on a portfolio’s long-term risk-return profile is nothing short of a miracle. While the benefits of adding just a few carefully chosen ingredients may not be immediately obvious, they compound meaningfully over time. A difference of just a few basis points per year can translate into substantial outperformance over multi-decade horizons. In this paper, the regime-adjusted strategy outperformed the static version by only 80 basis points annually, yet over nearly a century, that small edge resulted in doubling the cumulative return. In long-term investing, it’s prudent to seek every opportunity to enhance performance and reduce costs, even if the incremental gain seems negligible at first glance.

While this paper highlights the power of data-driven techniques in uncovering latent market patterns, the goal is not to replace human judgment with models, but to enhance it. Attempting to perfectly time the market is rarely wise, even understanding the present with clarity is a formidable challenge. The objective is to use models to help investors frame better questions, weigh probabilities more objectively, and make allocation choices that are informed, but not dictated, by data.

Finally, the limitation of the traditional 60/40 portfolio is evident when we extend our perspective beyond recent decades and investigate the full arc of financial history over the past century. These blind spots underscore the need for asset allocators to think both critically and creatively. In a world marked by rising uncertainty, clinging to conventional approaches is no longer sufficient. Standing on the shoulders of the giants who came before us, we need to remain open to new ideas, adopt the latest tools, and explore innovative frameworks that can help us more effectively navigate the complexities of today’s markets.

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